



BEST PRACTICE SERIES

Communicating VIMS Events Using Critical Event Advice

Application Maintenance Site Management Component Rebuild Safety MARC Management

Communicating VIMS Events Using Critical Event Advice0

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1.0 Introduction

Depending on the site, VIMS events can occur on machines occasionally, to very frequently. A system event indicates that the VIMS system has a fault, while a machine event indicates that the machine has been operated outside its intended design envelope. Managing VIMS events is critical to maintaining good machine health. This Best Practice describes how to communicate the occurrence of machine events to the customer, in a way that is proactive & helpful, rather than reactive and blaming. This helps make the Dealer/Customer relationship more positive, and increase customer satisfaction by taking the stance that “Prevention is better than Cure”. It also helps to identify and communicate operator practices that may not be safe.

The Critical Event Advice can lead to greater customer acceptance and commitment towards application management, a potential increase in component life, and the resulting reduction in machine operating costs. Prior to the Critical Event Advice being implemented onsite, the following difficulties were experienced:

- Customers saw VIMS events as a hitting stick for denying poor component life
- Customers were defensive about how the machines were operated
- It was difficult to convince customers to change operator behavior

2.0 Best Practice Description

Table 1 below describes how the Critical Event Advice (CEA) process works, from downloading the VIMS data, to how the customer should be approached.

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Table 1 – CEA Process Description

Task	Description	Person Responsible	When
1.	All contract machines will be downloaded at +/- 250 hour intervals.	MSS Coordinator Foreman	250 hour intervals
2.	Information will be uploaded into the site specific Equipment Health download spreadsheet.	Project Manager	At time of download
3.	All site specific spreadsheets will be copied to the dealer server.	Site Admin	At time of upload
4.	All Critical Event exceptions will be communicated to the customer using the Critical Event Advice format.	Project Manager	Within 24 hours of download
5.	Initial Communication to the customer MUST be face to face. Initial communication will be followed with an electronic copy of all documentation	Project Manager	Within 24 hours of download
6.	All recommendations will be followed up and or implemented	Project Manager	Within 14 days
7.	Summary of all Critical Events will be included in the Monthly Report using the Critical Event Summary format.	Project Manager	Monthly

NOTE: It is critical that the Dealer Project Manager approach the customer face to face initially (within 24 hours), to help personalize the process, and illustrate the importance of the information.

The first task is to download the VIMS data from the machine. The second task, as stated above, is to add the events to the Site Specific Equipment Health Spreadsheet. The purpose of this spreadsheet is to keep a running tab of the machine event history of each unit, and to provide the ability to trend events over time. Figure 2 below is an example of what this spreadsheet can look like.

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ET Logged Events

Unit 1

Model - 789

Serial No - #####

MODULE		ENGINE				STEER/BRAKE		TRANSMISSION			FUEL BURN			VIMS				
Download Date	Machine SMU Reading	Diagnostic Clock	Low Engine Oil Pressure Warning	High Crankcase Pressure Warning	Engine Overspeed Warning	Diagnostic Clock	Brake Stroke Hi	Diagnostic Clock	Coasting in Neutral Warning	Tranny Abuse Warning	Fuel Burn	Total fuel	Total Time	Diagnostic Clock	Transmission Abuse	Payload Overload more than 500m	Payload Overload less than 500m	Gear Select harmful
			E100 (1)	E101	E190		E01									3007	2917	2917
15/12/05	27727	6706	0	0	5	28156	0	8643	2	34								
08/02/06	28169	7148	0	0	7	28625	0	9111	3	34								
04/04/06	28904	7882	0	0	9	29371	0	9858	0	12								
04/05/06	29215	8192	0	0	10	29686	0	10173	0	24								
30/05/06	29517	8494	0	0	10	29996	0	10482	0	41								
18/01/07	30826	9805	0	0	11	11824	0	31338	0	52								

Figure 2 – Site Specific Equipment Health Spreadsheet

Task 4 calls for the communication of recent events to the customer, using the Critical Event Advice (CEA) format. All details of the event are spelled out in this document, including potential causes, potential ramifications, and suggested action items. Supporting documentation is also attached electronically when delivered via email. By going further than simply identifying the event, the communication becomes more proactive, arming the customer with the knowledge of what to do about it. The following page shows an example of an actual CEA.

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DATE
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Hastings Deering



CRITICAL EVENT ADVICE

REPORTED BY: Project Manager **UNITS AFFECTED:** Unit 1, Unit 2, Unit 3 **DATE:** 17.10.06

MACHINE MODEL: 24H **SERIAL NO:** 7KK **SMU:** 10000

CUSTOMER: ABC Mining

EVENT:

The mine site 24H Motor Grader Fleet experiences an unacceptably high number of engine overspeeds.

EVENT DESCRIPTION:

An engine overspeed is captured and logged each time the engine speed exceeds 2500 RPM

CONCLUSIONS: (What "CAUSED" the Event?)

Discussions with operators and previous transmission failure suggest that most likely cause of engine overspeed is high speed downhill travel with incorrect gear selected.

POTENTIAL RAMIFICATIONS:

- Significant to catastrophic engine failure
- Major power train failure (Already suffered on Unit 1)
- Premature brake failure
- Premature tandem chain failure

RECOMMENDATIONS:

- Operator refresher training would be the most effective measure to prevent further engine overspeeds
- Inspect valve train, con rod & main bearings on Unit 1

SUPPORTING DOCUMENTATION:

- Powerpoint presentation incorporating extracts from the mine site Equipment Health download spreadsheet.
- Event Log downloads from effected equipment
- Copy of correspondence indicating poor operating practice

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The final step in the process (task 7) is to provide the customer a summary of the events that have occurred over an extended period (monthly would be recommended). The following is an example of a monthly CEA summary.



**CRITICAL EVENT ADVICE
Monthly Report Summary (Example)**

“Name of Your Site”

Critical Event Advice			
Unit ID	Event	Quantity	Reported On
Truck 001	Engine Overspeed	7 @ 2200 RPM	20.08.06
		2 @ 2400 RPM	20.08.06
		1 @ 2600 RPM	20.08.06
	Brake Overheat	3	20.08.06
	Transmission Abuse	3	03.09.06
	Gear Select	7	03.09.06
Wheel Loader 002	As Above	As Above	As Above
Dozer 003	As Above	As Above	As Above
Grader 004	As Above	As Above	As Above

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3.0 Implementation Steps

A process for downloading VIMS data on a regular basis is necessary. Storage locations for the data will also be required, depending on data retention requirements. The customer may wish to have a copy on a local site server, and you may also wish to have a backup copy on a dealer server, to provide access to both parties.

Once data is available, use a template such as the one shown to record all details required about the event. Make an educated guess (based on machine and local mine site knowledge) about the cause of the event. Seek advice if necessary, regarding the potential ramifications of the event. Finally, suggest some actions that may help to reduce the occurrence in future.

It will also be necessary to establish a channel for communicating the CEA. The recipient may depend upon what type of event has occurred (e.g. Truck overloading events may need to be directed to the production superintendent, while low engine oil events may need to go to the Maintenance Manager or Planner). Whoever the intended recipient(s), they will need to be aware of the process, and support it.

Steps required to implement this Best Practice:

- Establish who will be involved (downloading, analyzing, receiving), and get their commitment
- Make sure there is a regular process for VIMS downloads and storage where it is accessible to everyone involved

4.0 Benefits

Increasing Quality (quality of service) – As a result of the CEA process, the dealer has been able to develop a much healthier relationship with both the maintenance and production departments onsite. The customer now sees the link between machine events, and increased downtime/costs.

Saving Time – Less time and effort is needed to rally around the customer, in order to effect change in operational practices, due to their heightened awareness.

Lowering Cost/Ton (through better component life) – With effective communication between departments, practices that are detrimental to component life can be managed more effectively.

Improving Operator/Technician Safety (identifying unsafe operating techniques) – By catching poor operating practices early, any potential safety issues can be addressed proactively, before an incident occurs.

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5.0 Resources Required

People

- Personnel to download VIMS data (if not automated)
- Project Manager to produce CEA report from template (minimal time required)

6.0 Supporting Attachments / References

Equipment Health Process Overview – PowerPoint presentation (Click on the “Attachments” tab within this document to view)



7.0 Related Best Practices

8.0 Acknowledgements

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